

5.16 (a) The effect of the soft iron can be modeled by a magnetic potential, which is the solution to the Laplace equation, $\nabla^2 \Phi_m = 0$. Due to the spherical symmetry, we can write the magnetic potential as $\Phi_m = \sum_{l=0}^{\infty} (A_l r^l + B_l / r^{l+1}) P_l(\cos \theta)$. We will be mainly interested in the solution in the cavity, and only need to retain the first term. Then, the additional magnetic field from the presence of the soft iron is

$$\begin{aligned} \vec{H}_1 &= -\nabla \Phi_m = -\left(\vec{e}_r \frac{\partial \Phi_m}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial \Phi_m}{\partial \theta} \right) \\ &= -\vec{e}_r \sum_{l=0}^{\infty} l A_l r^{l-1} P_l(\cos \theta) - \vec{e}_\theta \sum_{l=0}^{\infty} A_l r^{l-1} \frac{dP_l(\cos \theta)}{d\theta} \\ &= -\vec{e}_r \sum_{l=0}^{\infty} (l+1) A_{l+1} r^l P_{l+1}(\cos \theta) - \vec{e}_\theta \sum_{l=0}^{\infty} A_{l+1} r^l P'_{l+1}(\cos \theta) \end{aligned}$$

where we have used the fact that $dP_l(\cos \theta)/d\theta = P'_l(\cos \theta)$, $l \geq 1$. The total magnetic field inside the cavity is $\vec{H} = \vec{H}_0 + \vec{H}_1$, where $\vec{H}_0$ is the magnetic field from a current loop in the free space. We require $H_\theta = 0$ at $r = b$, or

$$\begin{aligned} -\frac{I a^2}{4} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n (n+1)!} \frac{1}{b^3} \left(\frac{a}{b} \right)^{2n} P'_{2n+1}(\cos \theta) \\ - \sum_{l=0}^{\infty} A_{l+1} b^l P'_{l+1}(\cos \theta) = 0, \end{aligned}$$

which leads to $A_{2n} = 0$, and

$$A_{2n+1} = -\frac{I a^2}{4} \frac{(-1)^n (2n+1)!!}{2^n (n+1)!} \frac{1}{b^{2n+3}} \left(\frac{a}{b} \right)^{2n}.$$

The radial magnetic field becomes

$$H_r = \frac{I a}{2r} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{r_2^{2n+1}}{r_1^{2n+2}} P_{2n+1}(\cos \theta) + \frac{I a^2}{4} \sum_{n=0}^{\infty} (2n+1) \frac{(-1)^n (2n+1)!!}{2^n (n+1)!} \frac{r^{2n}}{b^{2n+3}} \left(\frac{a}{b} \right)^{2n} P_{2n+1}(\cos \theta).$$

At the center of the loop, $r_2 = r = 0$, and $r_1 = a$, $\theta = 0$ we have

$$H_r(r=0, \theta=0) = \frac{I}{2a} + \frac{I}{4} \frac{a^2}{b^3} = \frac{I}{2a} \left(1 + \frac{a^3}{2b^3} \right), \quad H_\theta(r=0, \theta=0) = 0,$$

and $\frac{I}{2a}$ is the magnetic field at the center of the current loop. Therefore, we can see a $(1 + a^3/2b^3)$ enhancement in the presence of the soft iron.

(b) The radial magnetic field from the wire is

$$H_r = \frac{Ia^2}{4} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} (2n+1) \frac{r^{2n}}{b^{2n+3}} \left(\frac{a}{b}\right)^{2n} P_{2n+1}(\cos\theta)$$

By analogy, the magnetic field from a current loop is given by

$$H_r = \frac{I}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{r^{2n}}{a^{2n+1}} P_{2n+1}(\cos\theta).$$

for $r < a$. Therefore, by comparison, there is no image current can generate the desired magnetic field. However, for $b \gg a$, retain only the lowest order term, $n=0$, we can see that the

image current has a radius of $2b^3/a^2$.